

# **PROJECT 2**

**DATA CLASSIFICATION USING AI**

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DECODELABS**

# OVERVIEW

The objective is to create a machine learning model that can classify data into different categories. The model learns from labeled examples (training data) and predicts the correct category for new data.



# WORKFLOW

1. Load Dataset.
2. Preprocess Data.
3. Split Train/Test.
4. Train Model.
5. Predict.
6. Evaluate Accuracy



# TOOLS AND TECHNOLOGY

1. Python
2. Pyroid 3
3. Pandas
4. Scikit-learn
5. Matplotlib



# APPLICATION

- 1.Email spam detection
- 2.Disease Diagnosis.
- 3.Fraud detection.
- 4.Image classification.
- 5.customer segmentation
- 6.Face recognition.

# ALGORITHM

- 1.Import the required libraries.
- 2.Load the Iris dataset.
- 3.split the dataset into training and testing data.
- 4.Train the decision tree classifier.
- 5.Predict the test data.
- 6.Calculate the model accuracy.
- 7.Display the prediction and accuracy.
- 8.End the program.

# IMPLEMENTATION

The project was implemented using Python and the Scikit-learn library. First, the Iris dataset was loaded into the program and explored to understand its features and target classes. The dataset was then divided into training data (80%) and testing data (20%).

A Decision Tree Classifier was chosen as the machine learning algorithm because it is simple and effective for classification tasks. The model was trained using the training dataset so that it could learn the patterns in the data.

After training, the model was tested using the testing dataset to predict the flower categories. Finally, the model's performance was evaluated by calculating its accuracy, which showed that the model was able to classify the data correctly with high accuracy.



```

1 print("=====")
2 print("Project 2: Data Classification Using AI")
3 print("Internship: DecodeLabs")
4 print("=====")
5
6 # Import required libraries
7 import pandas as pd
8 from sklearn.datasets import load_iris
9 from sklearn.model_selection import
10 train_test_split
11 from sklearn.tree import DecisionTreeClassifier
12 from sklearn.metrics import accuracy_score
13
14 # Step 1: Load the Iris dataset
15 iris = load_iris()
16
17 # Create a DataFrame
18 data = pd.DataFrame(iris.data, columns=iris.
19 feature_names)
20 data["Flower Type"] = iris.target
21
22 # Display first five records
23 print("First Five Records:")
24 print(data.head())
25
26 # Display dataset information
27 print("\nDataset Shape:", data.shape)
28 print("\nMissing Values:")
29 print(data.isnull().sum())
30
31 # Step 2: Define Features and Target
32 X = iris.data
33 y = iris.target
34
35 # Step 3: Split the dataset
36 X_train, X_test, y_train, y_test = train_test_split(

```

# SOURCE CODE



```
35     X, y, test_size=0.2, random_state=42
36 )
37
38 # Step 4: Create the Decision Tree model
39 model =
40     DecisionTreeClassifier(random_state=42)
41
42 # Step 5: Train the model
43 model.fit(X_train, y_train)
44
45 # Step 6: Make predictions
46 predictions = model.predict(X_test)
47
48 # Step 7: Calculate accuracy
49 accuracy = accuracy_score(y_test, predictions)
50
51 print("\nModel Accuracy:", round((accuracy * 100,
52 2), "%"))
53
54 # Step 8: Predict a new flower
55 new_flower = [[5.1, 3.5, 1.4, 0.2]]
56
57 prediction = model.predict(new_flower)
58
59 print("\nNew Flower Details:")
60 print("Sepal Length =", new_flower[0][0])
61 print("Sepal Width =", new_flower[0][1])
62 print("Petal Length =", new_flower[0][2])
63 print("Petal Width =", new_flower[0][3])
64
65 print("\nPredicted Flower Name:", iris.
66     target_names[prediction[0]])
67
68 print("\nProject Completed Successfully!")
```

## SOURCE CODE

First Five Records:

| sepal length (cm) | sepal width (cm) | petal length (cm) | petal width (cm) | Flower |
|-------------------|------------------|-------------------|------------------|--------|
| 5.1               | 3.5              | 1.4               | 0.2              | setosa |
| 4.9               | 3.0              | 1.4               | 0.2              | setosa |
| 4.7               | 3.2              | 1.3               | 0.2              | setosa |
| 4.6               | 3.1              | 1.5               | 0.2              | setosa |
| 5.0               | 3.6              | 1.4               | 0.2              | setosa |

[5 rows x 5 columns]

Dataset Shape: (150, 5)

Missing Values:

sepal length (cm) 0  
sepal width (cm) 0  
petal length (cm) 0  
petal width (cm) 0  
Flower Type 0  
dtype: int64

Model Accuracy: 100.0 %

New Flower Details:

Sepal Length = 5.1  
Sepal Width = 3.5  
Petal Length = 1.4  
Petal Width = 0.2

Predicted Flower Name: setosa

Project Completed Successfully!

[Program finished]

# RESULT

1.Dataset loaded successfully

2.Model trained successfully

3.Accurate predictions were made

4.Achieved high accuracy(96-100%)

5.Project completed successfully



# CONCLUSION

The project successfully demonstrated the basic concepts of Artificial Intelligence and Supervised Machine Learning. The model was able to learn from the training data, make predictions on new data, and achieve high accuracy. This project helped in understanding data classification, model training, testing, and performance evaluation.

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A soft, circular watercolor splash in shades of light green and teal serves as a background for the text. It has a textured, painterly appearance with some darker green speckles and a few small dots trailing off to the bottom right.

thankyou